Diffusion Bridge Generation of Visible Light-to-Infrared Images Based on Multiscale Enhancement From Aerial Photography Perspective

Xin Wang, Wei Cai, Yao Ding, Xingyu Di, Zhongjie Yin, and Haoran Jia

***Abstract*—To address the challenges of imbalanced multiscale feature rhythm, network representation defects, and insufficient loss constraints in the generation of infrared images converted from aerial visible light images, a diffusion bridge generation model focusing on multiscale enhancement is proposed. Starting from the full-link design of multiscale generation, the model, in terms of diffusion process scheduling, designs a multiscale scheduling optimized differential equation, adjusts multiscale features by combining with a gated polynomial enhancement term, and realizes time-varying variance scheduling with a thermodynamic factor, so as to adapt to the logic of first constructing large-scale structures and then refining small-scale thermal details. In terms of network structure enhancement, it builds a time-modulated multiscale enhancement network (MEN), which regulates the intensity of infrared thermal features at each scale according to time steps—ensuring the thermal distribution of large-scale structures in the early stage and improving the authenticity of small-scale details in the later stage; it also proposes a frequency decomposition and fusion module, which decouples high-frequency and low-frequency features through discrete wavelet transform (DWT), repairs high-frequency details by virtue of discrete Fourier transform, and realizes dynamic fusion by combining time parameters. In terms of the loss function, it uses high-, medium-, and low-frequency losses to constrain pixel accuracy, and matches LPPIS perceptual loss to optimize semantic consistency, thus constructing an information closed loop. In the comprehensive scene test on the DroneVehicle dataset, the model outperforms 14 mainstream generative adversarial network (GAN) and diffusion models, with core indicators as follows: peak signal-to-noise ratio (PSNR) of 15.06, structural similarity (SSIM) of 49.78, learning-perceived image block similarity (LPIPS) of 30.71, and Frechet inception distance (FID) of 43.88; it also shows outstanding performance in subscenarios—for single-target scenarios, PSNR is 16.78 and SSIM is 59.48; for dense-target scenarios, FID is 34.67; for mixed-target scenarios, compared with the second-best method, PSNR increases by 0.58% and FID decreases by 15.36%. The model successfully distinguishes cross-category multiscale features and realizes the generation of multiscale infrared images in complex aerial scenarios.**

***Index Terms*—Aerial, deep learning, diffusion bridge, multiscale enhancement, visible-to-infrared image generation.**

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I. Introduction

With the development of aerial photography technology, tasks such as target detection for objects using images captured by aerial camera payloads have become increasingly mature. Infrared images have the significant advantage of being less constrained by weather and lighting conditions, enabling them to achieve good imaging results and anti-interference capabilities in low-light scenarios such as foggy days, rainy days, and nights [1]. Therefore, aerial infrared images play an irreplaceable role in fields such as military, agriculture, and commerce [2], [3], [4], [5]. However, compared with widely used visible light cameras, the cost of infrared imaging equipment remains relatively high. Consequently, it is of great importance to realize the generation of aerial infrared images using the most common form of visible light images.

A great number of attempts have been made by existing methods to generate infrared images from visible light images. The most basic approach is to use image processing technology based on physical models [6], [7]. This method addresses the issue of creating something from nothing, but it relies on physical formulas and thus struggles to adapt to complex environments. With the rise of deep learning technology, it has become feasible to achieve cross-domain conversion from visible light images to infrared images using neural networks. Generative adversarial networks (GANs for short) [8], [9] have been proven to be effective tools for learning such mappings. By leveraging the idea of generative adversarial learning, a variety of image-to-image conversion methods have emerged, enabling effective mapping between images. For instance, CycleGAN [10] utilizes cycle consistency loss to impose structural consistency constraints on network generation. EGGAN [11] enhances the object boundary information of generated images by means of edge enhancement constraints. DRAVIT [12] achieves diverse and realistic conversion from aerial visible light images to infrared images through the integration of geometric consistency constraints. Wu et al. [13] proposed an innovative model that skillfully employs

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steganography technology to actively suppress the generation of false features in generated images during the feature learning process, thereby effectively enhancing the semantic consistency of cross-generated images. Although GAN-based generation methods exhibit good conversion performance, they are prone to the problem of mode collapse during training [14], [15]. Recently, compared with GAN-based models, diffusion models have shown competitiveness in image generation. Conditional diffusion models [16], [17], [18], [19], [20] have been applied to image-to-image translation tasks. This method treats the source image in image-to-image conversion as conditional image generation and gradually generates the target image from noise through a diffusion process, overcoming the mode collapse problem of GAN models. Li et al. [21] proposed the Brownian diffusion bridge method (BBDM), which improves the quality of image generation and spatial consistency by utilizing the spatial alignment information of paired data. In 2024, scholars such as Kim et al. [22] proposed the unpaired neural Schrödinger bridge (UNSB) method. Its core lies in formulating the Schrödinger bridge problem as a series of adversarial learning problems. Through this transformation, advanced discriminators and regularization techniques can be integrated to efficiently learn the Schrödinger Bridge between unpaired data. Ran et al. [23] proposed DiffV2IR to achieve semantic-aware conversion. Through this new image conversion framework, unified visual language understanding is realized, and the generation information of infrared images is enhanced using data from text language and binary mask maps.

However, most of the aforementioned methods represented by BBDM [21] focus on datasets from the eye-level perspective, and have little involvement in generating infrared images from visible light images based on the aerial photography perspective. Especially from this perspective, the scales of vehicles, buildings, woods, roads, and so on, vary; moreover, different vehicles present different scales in different scenarios, leading to the multiscale problem in image generation. Although DRAVIT [12] targets images based on the aerial photography perspective and achieves relatively good results, the images involved are relatively clear, and its training dataset is divided into daytime and nighttime datasets, without covering adverse lighting conditions such as low light and fog. Meanwhile, due to the lack of corresponding text data and binary mask maps in the current paired datasets of aerial visible light and infrared images, it is temporarily impossible to use the DiffV2IR [23] method to realize the generation of aerial infrared images. Therefore, in view of the current multiscale problem under multiscenario complex conditions, we propose a diffusion bridge generation model for visible light-to-infrared images based on multiscale enhancement, which specifically addresses the multiscale problem under multiscenario complex conditions from the aerial photography perspective.

To achieve multiscale enhancement of infrared images generated from aerial images, we have made innovations in the following four aspects.

1) A differential equation for diffusion process scheduling is designed. A gated polynomial enhancement term is introduced into the drift term of the diffusion differential equation, and a thermodynamic factor is embedded into the perturbation term to dynamically adjust the generation weights of multiscale features and variance scheduling. This mechanism solves the problem that traditional linear scheduling cannot match the law of multiscale generation, and lays a foundation for the orderly generation of multiscale features in the conversion from aerial visible light-to-infrared images.

2) An architecture is constructed, which integrates a preposed time-modulated multiscale enhancement network (MEN) and an embedded multiscale frequency decomposition and fusion module, to dynamically adjust the intensity of thermal features and decouple high-frequency and low-frequency features. This structure breaks through the bottlenecks of high-frequency and low-frequency interference as well as high-frequency detail loss in traditional networks, and realizes the collaborative optimization of structure, details, and scales.

3) The multiscale losses of high-, medium-, and low-frequencies are integrated with the LPPIS perceptual loss, forming a closed loop of forward generation and backward constraint together with the diffusion scheduling and network structure. This design avoids the loss of multiscale information and semantic deviation, and ensures the information consistency of the full-link generation.

4) Experiments show that the proposed method is comprehensively superior to the current state-of-the-art models and can generate high-fidelity infrared images. It can provide accurate data support for scenarios such as nighttime reconnaissance and harsh environment monitoring, reduce reliance on high-cost infrared equipment, and promote the wide application of aerial infrared technology.

II. Related Work

A. Generation Models of Infrared Images

As an important part of image generation in computer vision, visible light-to-infrared image generation has wide applications in military detection, autonomous driving, image matching, agricultural monitoring, and more. In the early 21st century, physical model-based generation methods, which are built on thermal radiation theory and optical transmission models, were proposed, enabling the physical generation of infrared images. Jiang et al. [24] adopted a hybrid approach to model infrared targets and backgrounds based on geometry and multispectral textures. However, relying solely on physical models, such generation methods require the acquisition of large amounts of accurate data, struggle to handle complex scenarios, and have limited generation quality. Entering the 2010s, with the development of deep learning technology, GANs flourished, bringing revolutionary progress to image generation.

Isola et al. [9] introduced the GAN model into image generation tasks and proposed the Pix2Pix framework, which realizes pixel-level mapping between input and output images. On this basis, Pix2PixHD [25] explored methods for generating higher-quality high-resolution images. CycleGAN [10], DiscoGAN [26], and DualGAN [27] utilize cycle consistency loss to realize the generation of unpaired images, solving the

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problem of limited paired datasets and ensuring structural consistency between source domain images and generated images. The MUNIT method [28] put forward the assumption of a shared latent space, which implicitly includes cycle consistency constraints. It leverages the shared relationships among multiple input images to achieve high-quality image generation. However, the cycle reconstruction process causes translated images to contain all texture and edge features from the source domain, which results in generated images incorporating features of source domain images and reduces the authenticity of target images. Therefore, to address the limitations of cycle consistency constraints, various methods have introduced new constraints [29], [30], [31], [32], [33]. ACLGAN [29] proposed adversarial consistency constraints to break the limitations of original cycle constraints; Lee et al. [30] captured cross-domain and domain-specific shared information through disentangled representation learning, significantly reducing model collapse. Chang et al. [32] used domain-specific mapping to remap latent features in the shared content space to domain-specific content spaces, enabling images to be encoded more appropriately for cross-domain generation. Lee et al. [11] realized edge-guided RGB-to-infrared image generation through the constraint of edge information.

Beyond the aforementioned optimizations targeting cycle consistency constraints, researchers have also made multifaceted improvements and innovations to image generation models from other dimensions. At the level of feature fusion and conditional guidance, multimodal conditional guidance can be uniformly encoded into 1-D feature vectors, which are then effectively concatenated and fused with features in the middle layers of the network [34], [35]. This design provides a unified feature expression form for different types of input conditions (such as images, text, and speech), facilitating information interaction and collaboration in cross-modal tasks.

Regarding the core challenges commonly existing in the training process of GAN models—including training instability, mode collapse, and difficulty in convergence, which severely restrict the upper limit of model performance and generalization ability—Huang et al. [36] proposed a high-performance regularized relativistic GAN loss function. Through rigorous mathematical analysis, they proved the local convergence of the model, effectively alleviating problems such as gradient vanishing, oscillation, or nonconvergence during training, and providing important support for the stable training of GANs. In addition, Shi et al. [37] proposed a new framework of "visual Gaussian quantization (VGQ)," which integrates 2-D Gaussian functions into traditional visual codebook quantization, significantly enhancing structural modeling capabilities; Tang et al. [38] proposed a GAN based on cross-attention to solve challenging character image generation tasks.

To achieve high-quality generation of aerial infrared images, Lu et al. [39] proposed the contour CycleGAN method. By bridging the spatial domain between visible light and infrared images, this method eliminates the domain gap between the two. For the purpose of realizing RGB-infrared cross-modal detection based on uncertainty-aware learning, Sun et al. [40] established a paired dataset of aerial images, providing a data foundation for image generation methods applied in AVIT. Based on feature disentanglement and cross-modal reconstruction, Li et al. [41] realized the generation of infrared images from remote sensing visible light images. The DRAVIT method [12] introduced disentangled representation learning, and achieved effective mapping between aerial visible light images and infrared images by combining geometric consistency constraints with a dual learning framework. However, all the aforementioned methods have the problem of multiscale generation in aerial perspective image generation, and are unable to adapt to the characteristic of inconsistent scales in infrared image generation existing in complex scenarios from the aerial perspective.

B. Image Generation Based on Diffusion Models

Drawing on the idea of diffusion processes in physical thermodynamics, diffusion models gradually add noise over time to learn data distribution, transforming from disorder to order. Unlike traditional GANs, diffusion models learn incrementally, making them less prone to mode collapse during training. Diffusion probabilistic models were first proposed by Bacciu et al. [42], which define a forward diffusion process and a reverse generation process to enable data generation. The forward process gradually adds noise to the data, while the reverse process gradually denoises by learning the distribution of noise, thereby generating the target data.

Ho et al. [43] proposed the denoising diffusion model, which attracted widespread attention. By optimizing the variational lower bound, this model significantly improved the generation quality of the model and expanded the application of diffusion models in image generation. The denoising diffusion probabilistic model (DDPM) generates images through gradual denoising: it predicts the noise at the current step and restores the image incrementally, possessing the ability to capture both the details and global structure of the image. Song et al. [44] proposed the denoising diffusion implicit model (DDIM), which uses a non-Markovian process sampling method to accelerate model sampling, reduce the number of model sampling steps, and improve training efficiency.

Batzolis et al. [17] proposed the conditional diffusion model, which incorporates conditional information into diffusion models. Since then, conditional diffusion models have received great attention, enabling the integration of information such as text, categories, and images, with continuous improvements in model efficiency and generation quality. Rombach et al. [45] proposed latent diffusion models (LDMs), which map high-resolution images to a low-dimensional latent space and perform the diffusion process in this latent space. This greatly reduces computational costs and enables efficient and high-quality image generation.

Li et al. [21] introduced the Brownian bridge into diffusion models for the first time and proposed the BBDM. This model differs significantly from previous diffusion models: the two ends of its diffusion bridge are anchored to the source image and the target image, respectively. Similar to the modeling logic of GANs, this design reduces the difficulty of constructing the mapping relationship between the source domain

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and the target domain in image generation, can significantly improve the generation quality of target images, and avoids the decline in model training generalization and stability caused by excessive interdomain differences.

Kim et al. [22] proposed the UNSB, which enables the Schrödinger bridge to realize the conversion between two arbitrary distributions and can combine advanced discriminators and regularization to learn the correlation information of unpaired data. Ran et al. [23] introduced the DiffV2IR framework to achieve semantic-aware conversion. Through unified visual language understanding, it enhances the information dimension of infrared image generation using text and binary mask map data. However, the existing datasets lack the text and binary mask map data required by DiffV2IR, resulting in this method being unable to effectively improve the generation ability of aerial infrared images.

Overall, the aforementioned methods, including BBDM [21], UNSB [22], and DiffV2IR [23], all lack adaptability to multiscale features in the task of generating infrared images from aerial visible light images. Therefore, this article proposes a diffusion bridge generation model for visible light-to-infrared images based on multiscale enhancement from the aerial photography perspective, aiming to achieve effective capture and conversion of multiscale features during the generation process.

C. Basic Theory of Diffusion Models

1) Foundations of the Diffusion Model: The forward diffusion process can be expressed as

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = N\left(\boldsymbol{x}_t;\ \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I}\right) \tag{1}$$

where $\boldsymbol{x}_t$ is the noise image at step t, β_t is the noise variance, and N represents the Gaussian distribution.

The backward diffusion process progressively denoises the noise distribution by learning it

$$p_\theta(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = N\left(\boldsymbol{x}_{t-1}; \mu_\theta(\boldsymbol{x}_t, t), \Sigma_\theta(\boldsymbol{x}_t, t)\right) \tag{2}$$

where μ_θ and Σ_θ are the mean and variance of the neural network prediction.

2) Fundamental Concepts of Brownian Bridge: A Brownian bridge is a conditional stochastic process with a fixed start and endpoint and intermediate paths obeying Brownian motion. Mathematically, a Brownian bridge can be defined as

$$B_t = W_t - tW_1, \quad t \in [0,1]. \tag{3}$$

B_t is the standard Brownian motion and t is the time endpoint. The Brownian bridge satisfies the following boundary conditions:

$$B_0 = 0, \quad B_t = 0. \tag{4}$$

In the image generation task, the start and end points of the Brownian bridge can correspond to the input image and the target image, respectively, and the intermediate path describes the generation process from the input to the target.

3) Brownian Bridge-Based Diffusion Model: In BBDM, the Forward Diffusion Process Is:

$$q(x_t \mid x_0, x_T) = N(x_t; \mu_t, \Sigma_t) \tag{5}$$

where μ_t and Σ_t are the mean and variance calculated from the properties of the Brownian bridge. Specifically,

$$\mu_t = x_0 + T_t(x_T - x_0) \tag{6}$$

$$\Sigma_t = \frac{t(T-t)}{T} I. \tag{7}$$

Specifically, in the actual diffusion process of the model, the intermediate process by μ_t can be expressed by the diffusion coefficient m_t, x_0, and x_T, that is, $x_t = m_t x_T + (1 - m_t)x_0$. Among them, m_t can be arranged by time t, and its formula is as follows:

$$m_t = m_{min} + t \cdot \frac{m_{max} - m_{min}}{T} \tag{8}$$

where m_t denotes the diffusion coefficient at time-step t, m_{min} and m_{max} are the minimum and maximum values of the diffusion coefficient, respectively, and T is the total number of time steps.

The inverse generation process progressively denoises the noise distribution by learning it

$$p_\theta(x_{t-1} \mid x_t, x_0, x_T) = N\left(x_{t-1}; \mu_\theta(x_t, t, x_0, x_T), \Sigma_\theta(x_t, t, x_0, x_T)\right) \tag{9}$$

where μ_θ and Σ_θ are the mean and variance of the neural network prediction.

The training objective of BBDM is to minimize the KL scatter between the inverse process and the true distribution

$$L = E_q(x_0, x_T)\left[\sum_{t=1}^{T} D_{KL}\left(q(x_{t-1} \mid x_t, x_0, x_T) \parallel p_\theta(x_{t-1} \mid x_t, x_0, x_T)\right)\right]. \tag{10}$$

By optimizing this objective, BBDM is able to learn the mapping relationship from the input image to the target image.

In the sampling process, BBDM starts from the noisy image and gradually denoises the target image. The specific steps are as follows.

1) Initialize to a noisy image.

2) For $t = T, T-1, \ldots, 1$, perform the following steps: calculate the mean $\mu_\theta(x_t, t, x_0, x_T)$ and variance $\Sigma_\theta(x_t, t, x_0, x_T)$.

3) Sampling from a Gaussian distribution x_{t-1}: $N(\mu_\theta(x_t, t, x_0, x_T), \Sigma_\theta(x_t, t, x_0, x_T))$.

4) Finalize the target image.

III. Methodology

To address the core issues of imbalanced multiscale feature generation rhythm, network representation defects, and insufficient loss constraints in existing models for visible light-to-infrared image generation from the aerial photography perspective, this article proposes a diffusion bridge generation model for visible light-to-infrared images based on multiscale enhancement, whose architecture is shown in Fig. 1. Existing technologies have significant limitations: among them, the

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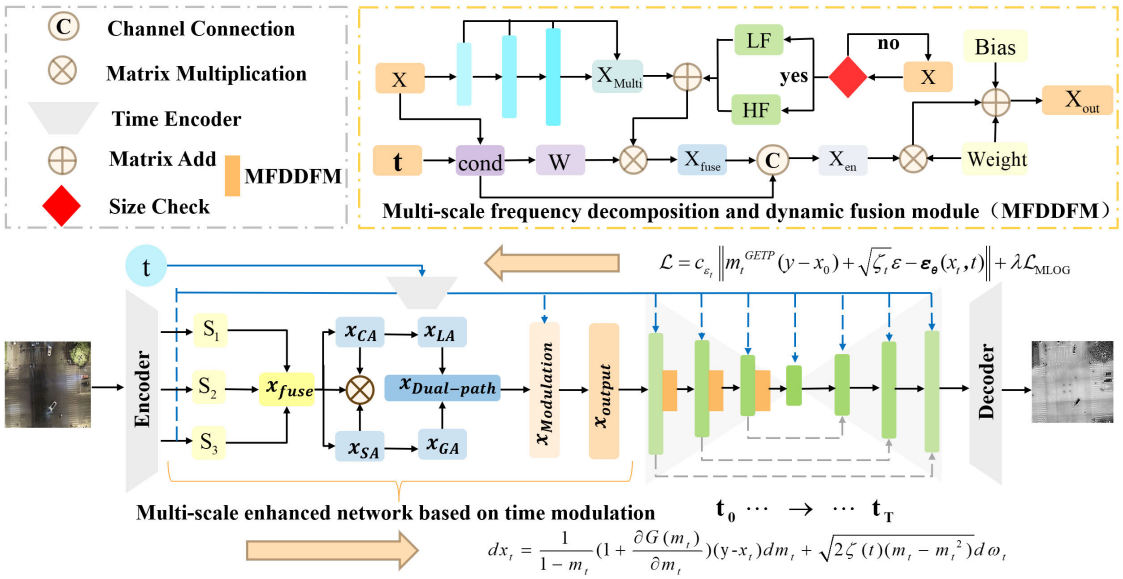


Fig. 1. Diffusion bridge generation of visible light-to-infrared images based on multiscale enhancement from an aerial photography perspective.

linear time-step scheduling of the Brownian bridge model cannot adapt to the multiscale generation logic of "first constructing large-scale structures and then refining small-scale details," leading to an imbalance between structure and detail generation.

Meanwhile, traditional denoising networks lack a multiscale feature disentanglement mechanism, which causes problems such as deep loss of high-frequency details, representation conflicts between large and small scales, and unbalanced frequency distribution. Furthermore, most loss functions are either single-pixel loss or perceptual loss, and do not impose targeted constraints on multiscale features of high, medium, and low frequencies, resulting in deviations in multiscale semantic consistency.

To tackle the above problems, this article constructs a full-link multiscale enhancement mechanism by combining the modules in the figure from three dimensions: diffusion process scheduling, network structure enhancement, and loss function constraint. In terms of diffusion process scheduling, a gated polynomial enhancement term and a thermodynamic factor are introduced into the Brownian bridge diffusion differential equation; the gated polynomial enhancement term dynamically adjusts the multiscale feature generation weights at different time steps through nonlinear modulation, and the thermodynamic factor adapts to the learning needs of infrared multiscale features with time-varying variance, enabling the diffusion process to actively conform to the multiscale generation rhythm of "first structure, then details." In terms of network structure enhancement, a time-modulation-based MEN (MENBOTM) is placed before the denoising network UNet; through the dynamic modulation of time steps by the time encoder in the figure, targeted enhancement of infrared thermal features at various scales under different time steps is achieved; at the same time, a multiscale frequency decomposition and dynamic fusion module (MFDDFM) is embedded in UNet, which iteratively decomposes high-frequency and low-frequency features via discrete wavelet transform (DWT), enhances high-frequency details through grouped convolution of discrete Fourier transform, and then combines channel connection and dynamic weight generation to solve multiscale representation defects (including high-frequency detail loss, representation conflicts between large and small scales, and unbalanced frequency distribution) from the frequency-domain dimension. In terms of loss function constraint, the original loss function formula is combined with multiscale feature losses of three frequency bands (high, medium, and low) and LPPIS perceptual loss to form a closed-loop enhancement of multiscale features, which ensures the full retention of multiscale information in gradient propagation and ultimately achieves multiscale consistency optimization across pixel-structure-semantic dimensions.

A. Scheduling Optimization Based on Multiscale Enhancement

To address the limitation of diffusion bridge models in multiscale feature generation, a scheduling optimization method based on multiscale enhancement is proposed, with two key improvements made in time-step parameter design and variance scheduling.

Equations (11) and (12) represent the differential equations based on the Brownian bridge model and our proposed model, respectively,

$$dx_t = \frac{\mathrm{y} - x_t}{1 - m_t} dm_t + \sqrt{2\left(m_t - m_t^2\right)} d\omega_t \tag{11}$$

$$dx_t = \frac{1}{1 - m_t}\left(1 + \frac{\partial G\left(m_t\right)}{\partial m_t}\right)(y - x_t)\, dm_t + \sqrt{2\zeta\left(t\right)\left(m_t - m_t^2\right)} d\omega_t. \tag{12}$$

Among them, dm_t denotes the drift term, which controls the image to move toward the target y; $d\omega_t$ denotes the diffusion term, which controls random perturbations. Compared with the original model, we add $(\partial G(m_t))/(\partial m_t)$ to the drift term,

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where $G(m_t)$ is the gated polynomial enhancement term. In the drift term, nonlinear enhancement is used to strengthen the multiscale effect. Meanwhile, the thermodynamic factor $\zeta(t)$ is added to the diffusion term, and thermodynamic adaptability is utilized to realize variance scheduling for multiscale feature generation.

1) Time-Step Parameter Optimization for Gated Polynomial Enhancement: To address the issue that the time-step parameters of existing Brownian bridges fail to be optimized for the formation stages of features at different scales, this study proposes a time-step parameter optimization improvement method with gated polynomial enhancement to enhance the multiscale feature generation capability and break the constraints of linear time steps on feature generation. By adding a gated polynomial enhancement term to the original time-step formula, the time-step scheduling is optimized; the time-step enhancement is adaptively adjusted according to the stage of the time step. The optimized formulas are shown in the following equation:

$$\beta_t = \frac{m_t}{1-m_t} + \sqrt{m_t(1-m_t)} \cdot \sigma\left(5^*(m_t - 0.5)\right) \tag{13}$$

$$m_t^{GETP} = \frac{\beta_t}{1-\beta_t}. \tag{14}$$

Among them, m_t is the time-step parameter of the original Brownian bridge, β_t is the optimized signal-to-noise ratio (SNR), m_t^{GETP} is the optimized time-step parameter, and σ is the sigmoid function. In (13), $(m_t)/(1-m_t)$ is the base term, $(m_t(1-m_t))^{1/2} \cdot \sigma(5^*(m_t - 0.5))$ is the gated enhancement term $G(m_t)$, and $\sigma(5^*(m_t - 0.5))$ is the gating factor, which controls the enhancement intensity of the enhancement term; its specific expression is shown in the following equation:

$$g(m_t) = \frac{1}{1 + e^{-5(m_t - 0.5)}}. \tag{15}$$

In the optimization of time-step parameters, the gating factor $g(m_t)$ simulates the thermal saturation characteristic and analogizes the Fermi–Dirac-like distribution in thermodynamics; β_t simulates the non-Fourier heat conduction effect, with $G(m_t)$ serving as the non-Fourier correction term to describe the Casimir heat flux limitation. By leveraging thermodynamics, it achieves precise gating of time-step parameters, thereby enhancing the multiscale characteristics of the network.

As shown in Fig. 2(a), through the gated enhancement term, this study can achieve three-stage feature optimization in the feature space: when m_t is less than 0.3, the system enters the early acceleration stage, where the growth rate of m_t increases to accelerate the formation of early stage features and establish the basic structure more quickly. This enhances the weight of the target signal to rapidly capture the contours and positions of objects, which is conducive to generating large-scale features, defining the general contours of objects and locating heat sources, so as to avoid excessive loss of large-scale structures and ensure the effectiveness of low-frequency features. When m_t is greater than 0.3 but less than 0.7, the growth rate of m_t slows down and the system enters the mid-term transition stage; here, the weight of the target signal is further increased, and the network focuses on the fine features and infrared characteristics of the target to realize the nonlinear conversion from visible light to thermal radiation. When m_t is greater than 0.7, the system enters the late stabilization stage, where the growth rate of m_t decreases and basically overlaps with the original curve; the natural attenuation mechanism protects fine features, performs fine-tuning on existing features, avoids noise caused by excessive enhancement, and maintains the details of thermal spots without generating artifacts.

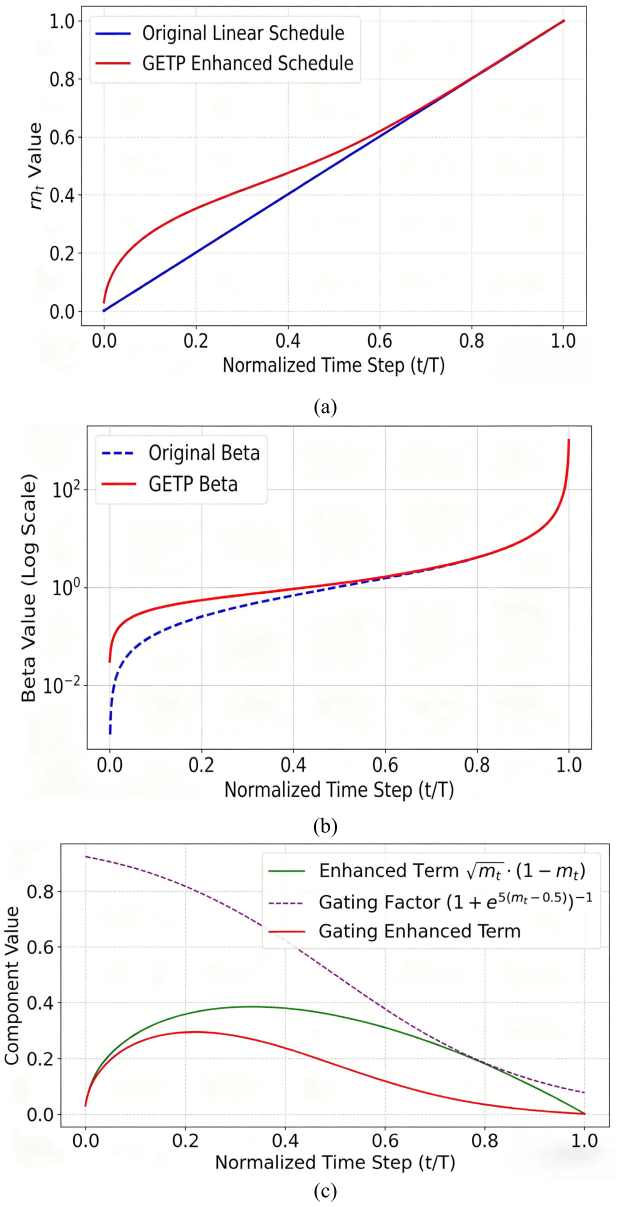


Fig. 2. Visual comparison between time-step parameter optimization with gated polynomial enhancement and the original method. (a) Three-stage feature optimization controlled by the gated enhancement term. (b) Logarithmic-scale SNR variation across the three generation stages. (c) Performance comparison of polynomial term, gating factor, and gated enhancement term.

Fig. 2(b) shows the logarithmic scale change of the SNR; it can be observed that in the early stage, the SNR increases super-linearly, which improves the injection intensity of the target signal and enhances the integrity of scale structures; in the mid-term, a stable transition is achieved to ensure the smooth diffusion from visible light to infrared; in the late stage, emphasis is placed on enhancement to improve the ability of generating high-frequency details.

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In Fig. 2(c), we can observe the comparison among three curves: the gating factor, the polynomial term, and the gated enhancement term. The values of the original polynomial term are excessively large in the early acceleration stage, mid-term transition stage, and late stabilization stage, which will lead to unstable generation of both large-scale structures and fine features in the early stage and cause excessive noise. Therefore, we introduce the gating factor; through the gating factor, the values in the early stage are reduced slightly, and the values in the mid-to-late stages are reduced significantly, thereby achieving precise control over the time-step parameters of the generation process.

2) Thermodynamics-Inspired Dynamic Variance Scheduling: To adapt variance scheduling to the multiscale feature generation of infrared images, inspired by thermodynamics, we propose a thermodynamics-inspired dynamic variance scheduling method. In this method, we introduce the idea of cosine annealing, adjusting the original static maximum variance value to a time-varying variance to solve the problem of unbalanced multiscale feature learning caused by the fixed maximum variance value. This time-varying variance is regulated by a thermodynamic factor, and an effective temperature field is used to simulate the change of time-varying variance in thermal diffusion during infrared image generation. The formulas are shown in the following equation:

$$\xi(t) = 0.5 \cdot \left[1 + \cos\left(\pi \frac{t}{T}\right)\right] \tag{16}$$

$$\delta_{max}^{dy}(t) = \delta_{max} \cdot (0.9 - 0.1 \cdot \xi(t)) \tag{17}$$

$$\delta_t = 2 * \left(m_t - m_t^2\right) * \ \delta_{max}^{dy}. \tag{18}$$

Equation (16) is the mathematical expression of the thermodynamic factor, which analogizes the thermal relaxation process. The cosine curve of $\xi(t)$ corresponds to the quantitative expression of the relaxation oscillation thermal saturation degree of $1 + \cos(\omega t)$ in the thermodynamic system, as shown in Fig. 3(a). In the initial state of the system, thermal disturbance is the smallest; in the middle stage, the curvature changes rapidly and heat exchange is the most active; in the later stage, the system reaches the neighborhood of the equilibrium state, the entropy production rate decreases significantly, and overall, the complex thermal response is converted into cosine parameterization.

Equation (17) is the mathematical expression of the time-varying variance, corresponding to the magnitude of thermal fluctuations in thermodynamics. It is matched with the time-step parameters of the gated polynomial enhancement; in the dynamic variance regulation, physical constraints are parameterized, and the variance is controlled between 0.8 and 0.9 to prevent noise oscillation caused by excessive variance changes. As shown in Fig. 3(a), the time-varying variance shows an overall upward trend: in the early stage of diffusion, when the input of the target signal is relatively small, a smaller time-varying variance helps maintain the initial stability of the generation process and preserve the large-scale structure of the input image. In the late stage of diffusion, when the target signal is relatively abundant, a larger time-varying variance can avoid excessive constraints, prevent the model from converging prematurely to a potentially unsatisfactory state,

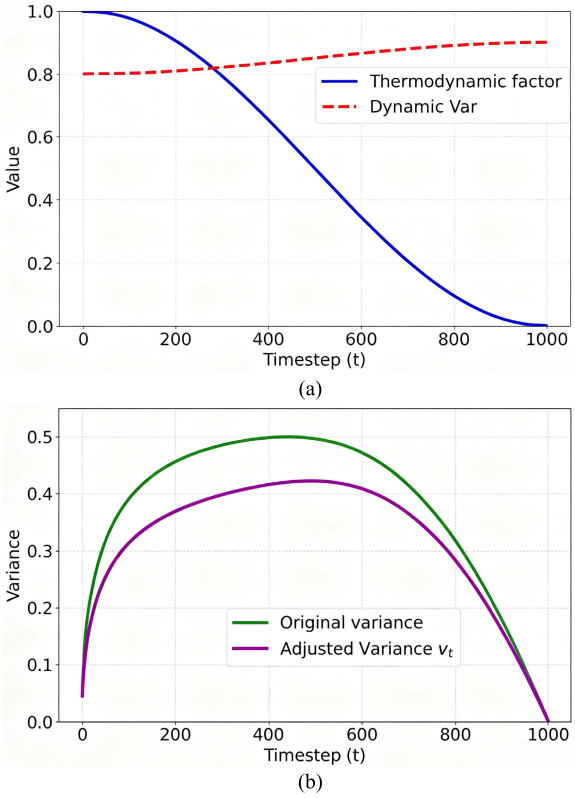


Fig. 3. Visual comparison between thermodynamics-inspired dynamic variance and the original method. (a) Time-varying variance curve and its physical implication for dynamic regulation. (b) Optimized variance curve and its multi-scale feature suppression effect.

and facilitate the generation of small-scale infrared image features with rich details. Equation (18) represents the form of the final variance formula; its curve is shown in Fig. 3(b). The overall level of variance is appropriately reduced, which ensures the generation quality of features at all scales and thereby constrains and optimizes the generation of infrared images.

B. MEN Based on Temporal Modulation

Since the thermal distribution features of infrared images under aerial photography perspectives exhibit significant heterogeneity across different scales, and during the generation process, the model needs to simultaneously ensure the integrity of the overall image structure and the authenticity of thermal distribution conforming to infrared physical laws at different time steps, existing networks suffer from issues such as unsynchronized generation of large-scale structures and enhancement of small-scale thermal details, as well as conflicts in multiscale thermal feature representation, due to the lack of a dynamic adaptation mechanism between temporal information and multiscale features. To address this, this study proposes an MEN based on temporal modulation. Through the architectural design of the temporal encoder, conditional modulation network (CNM), multiscale enhancement module (MEN), and other components shown in Fig. 4, the network realizes the encoding of temporal information and conditional modulation of multiscale features, aiming to achieve targeted enhancement of infrared thermal features at various scales under different time steps. Specifically, after the temporal

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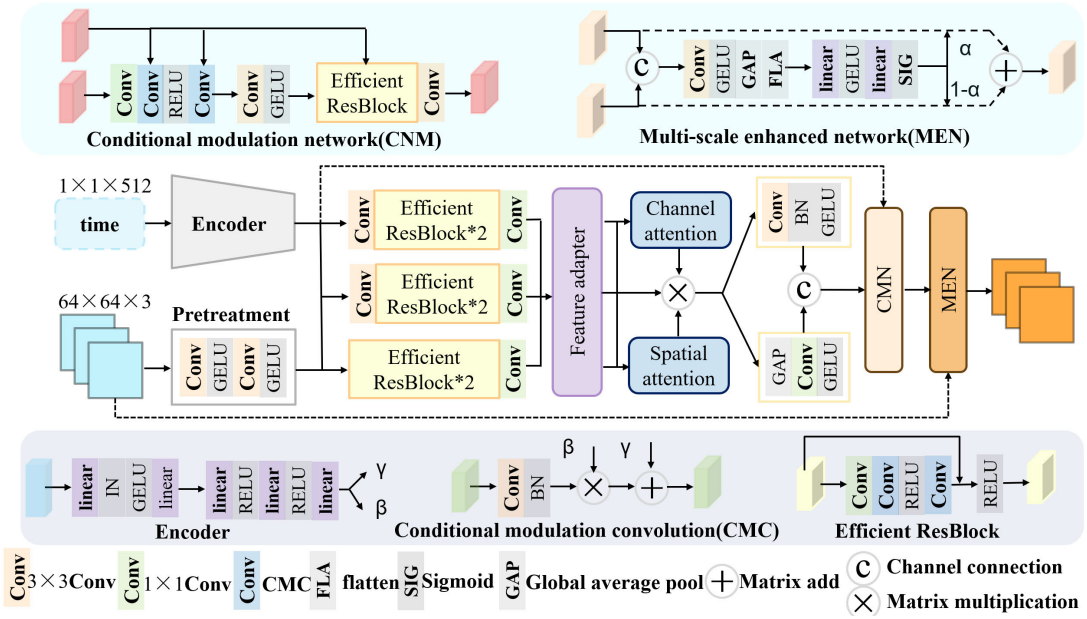


Fig. 4. Architecture diagram of MEN based on temporal modulation.

information is processed by the encoder, it conducts in-depth feature interaction with multiscale features through efficient ResBlocks in the CNM; subsequently, a feature adapter fuses the weighted information of channel attention and spatial attention to finely adjust the channel and spatial dimensions of multiscale features; finally, in the MEN, the weights of features at each scale are dynamically adjusted through channel connection and matrix operations. This enables the model to prioritize enhancing the rationality of thermal distribution for large-scale structures in the early stage of generation and accurately improve the authenticity of small-scale thermal details in the late stage of generation. Thereby, the overall image structure and infrared thermal distribution are coordinated and consistent across multiple time steps and scales, meeting the requirements of multiscale adaptation of structure-thermal features and infrared physical consistency in infrared image generation.

As shown in Fig. 4, its input consists of a temporal vector with dimensions of $1 \times 1 \times 512$ and a latent space image vector with dimensions of $64 \times 64 \times 3$.

First, to realize the modulation of multiscale features at different time steps, the input temporal vector is dimensionally reduced by an encoder, and parameters γ and β for conditional modulation are generated. Their mathematical expressions are shown in the following equation:

$$t_{feat} = linear\left(IN\left(GELU\left(linear\left(t\right)\right)\right)\right) \tag{19}$$

$$\left[\gamma,\beta\right] = linear\left(RELU\left(linear\left(RELU\left(linear\left(t_{feat}\right)\right)\right)\right)\right). \tag{20}$$

Equation (19) performs dimensionality reduction on the temporal vector t through two fully connected layers *linear*, reducing the dimension from 512 to 64 to form temporal features with the same dimension as the latent vector. The activation function *GELU* provides a smooth nonlinear transformation for dimensionality reduction, preserving more information, while the layer normalization *IN* stabilizes the training process and reduces internal covariate shift.

Equation (20) generates parameters for modulation based on the dimensionally reduced temporal features. It realizes the nonlinear transformation from features to parameters through three components *linear*, and effectively alleviates gradient issues using component *RELU*. The output parameters γ and β are used for feature scaling and bias adjustment, respectively, enabling dynamic modulation of features at all scales.

To achieve multiscale feature extraction, we utilize a multiscale extraction network composed of efficient ResBlocks, convolution layers, and feature adapters to extract features from the preprocessed latent image vectors, thereby realizing feature decomposition. Meanwhile, we use parameters γ and β output by the encoder for dynamic modulation. Its mathematical expression is shown in the following equation:

$$x_{fuse} = [S_1, S_2, S_3] = \bigcup_{i=1}^{3} f_i\left(\Gamma_i\left(Conv_i\left(x_{prep},\gamma,\beta\right)\right)\right). \tag{21}$$

Among them, S_1, S_2, and S_3 represent the features output by the scale branches without downsampling, $2\times$ downsampling, and $4\times$ downsampling, respectively, with channel dimensions of 64, 128, and 256. Specifically, S_1 is a small-scale feature, which is used to restore texture details, fill in edge details, and maintain the content structure; S_2 is a medium-scale feature, which is used to complement the local structure, reconstruct local consistency, and coordinate regional styles; and S_3 is a large-scale feature, which is used to maintain overall consistency and the coherence of global semantics, as well as control the overall style characteristics.

In (21), x_{fuse} denotes the preprocessed image vector; $Conv_i$, Γ_i, and f_i represent the downsampling convolution, efficient ResBlock, and upsampling convolution of the i branch, respectively; and $\cup$ stands for feature adaptation after upsampling, which forms a rich feature representation with 768 channels.

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To enhance the thermally sensitive regions of multiscale features, we design a dual collaborative attention mechanism and a dual-path attention mechanism after the extraction network, which are used to achieve feature calibration and feature integration for thermally sensitive regions. Among them, the dual collaborative attention mechanism consists of a channel attention mechanism and a spatial attention mechanism, and its mathematical expression is shown in the following equation:

$$\begin{aligned} x_{Collaboration} &= x_{fuse} \otimes \sigma\left(Conv_{1\times1}\left(GELU\left(Conv_{1\times1}\left(GAP\left(x_{fuse}\right)\right)\right)\right)\right) \\ &\quad \otimes \sigma\left(Conv_{3\times3}\left(x_{fuse}\right)\right). \end{aligned} \tag{22}$$

Among them, the second part of the formula represents the weight output by channel attention calculation: the input x_{fuse} undergoes adaptive average pooling to extract the global response value of each channel, and 1×1 convolution is used to reduce the number of parameters. After smoothing via the $GELU$ function, the dimension is restored to the original one, and the channel weight is output through the sigmoid function. The third part of the formula denotes the weight output by spatial attention calculation: 3×3 convolution is applied to extract local spatial relationships, and the sigmoid function is used to generate spatial attention weights. The generated weights are multiplied with x_{fuse} pixel-wise, so as to achieve the focusing and calibration of features in thermally sensitive regions in both channel and spatial dimensions through channel-spatial collaboration.

The dual-path attention mechanism consists of two parts: a local path and a global path. The local path preserves the high-frequency components and important texture details of $x_{Collaboration}$ while suppressing irrelevant responses; the global path provides noise-resistant context and maintains the overall structure. Its mathematical expression is shown in the following equation:

$$x_{Dual-path} = \begin{bmatrix} GELU\left(Conv_{1\times1}\left(GAP\left(x_{Collaboration}\right)\right)\right), \\ GELU\left(BN\left(Conv_{3\times3}\left(x_{Collaboration}\right)\right)\right) \end{bmatrix}. \tag{23}$$

The first part of the formula corresponds to the global path, which extracts global response values through GAP (adaptive average pooling) and outputs global features; the second part of the formula corresponds to the local path, which extracts local details of the calibrated feature $x_{Collaboration}$, outputs local features, and then adds these local features to the global features in the channel dimension to generate output $x_{Dual-path}$. This ultimately achieves feature integration, improves feature representation, and forms a high-level feature representation with spatial sensitivity and scale completeness.

To achieve effective full-process modulation of network features by time-step parameters, after module $x_{Dual-path}$, we design a CNM. Using the output parameters of the temporal encoder and conditional modulation convolution, we perform feature scaling and bias adjustment on $x_{Dual-path}$. After reprocessing the features, they are input into the combination of efficient ResBlock and 3×3 convolution to guide the diffusion generation process and further address the noise sensitivity issue. Its mathematical expression is shown in the following equation:

$$\begin{aligned} x_{Modulation} &= Conv_{3\times3}\left(\Gamma\left(GELU\left(Conv_{3\times3}\right.\right.\right. \\ &\quad \left.\left.\left.\left(Conv_{CMC}\left(RELU\left(Conv_{CMC}\right.\right.\right.\right.\right. \\ &\quad \left.\left.\left.\left.\left.\left(Conv_{1\times1}\left(x_{Dual-path}\right)\right)\right)\right)\right)\right)\right). \end{aligned} \tag{24}$$

Among them, $Conv_{CMC}$ refers to the conditional modulation convolution, and Γ denotes the efficient ResBlock function.

To balance the relationship between visible light structure and multiscale features, ensure the rationality and structural consistency of the generated images, and prevent information loss in multiscale feature extraction, we design a joint adjustment factor network after module $x_{Modulation}$. By jointly processing inputs x_{prep} and $x_{Modulation}$, the network generates an adjustment factor α, which is used to adjust the feature relationship between x_{prep} and $x_{Modulation}$ and output the optimal mixed result. Their mathematical expressions are shown in the following equations:

$$x_{Factor} = \sigma\left(GAP\left(GELU\left(Conv_{3\times3}\left[x_{prep}, x_{Modulation}\right]\right)\right)\right) \tag{25}$$

$$\alpha = \sigma\left(linear\left(GELU\left(linear\left(x_{Factor}\right)\right)\right)\right). \tag{26}$$

In (25), we use a combination of 3×3 convolution, $GAP, GELU$, and the sigmoid function to perform in-depth feature extraction on the input x_{prep} and $x_{Modulation}$, and output the feature weight x_{Factor}. Then, in (26), a combination of $linear$, GAP, and the sigmoid function is used to perform a nonlinear transformation on x_{Factor}, and output the adjustment factor α. Through this factor, the original input and multiscale results are mixed, and the final feature x_{output} is output, so as to achieve a balance between the structure and multiscale thermal characteristics.

C. Multiscale Frequency Decomposition and Dynamic Fusion Module

To address the multiscale representation defects of the denoising network in diffusion models during the generation of infrared images from visible light under aerial photography perspectives—including deep loss of high-frequency details, mutual interference of thermal features between large and small targets in a unified space, and unbalanced frequency distribution—this study proposes an MFDDFM. As shown in Fig. 5, the input feature X is subjected to multiscale frequency decomposition via DWT. After judgment by Size check, the feature is separated into low-frequency (L-Portion) and high-frequency (H-Portion) components, enabling decoupled representation of large and small targets in different frequency spaces and fundamentally avoiding mutual interference between low-frequency and high-frequency features. Subsequently, a low-frequency processor is applied to the low-frequency component to enhance the integrity of large-scale structures, while a high-frequency processor is used for the high-frequency component—combining discrete cosine transform (DCT), channel attention, and spatial attention—to

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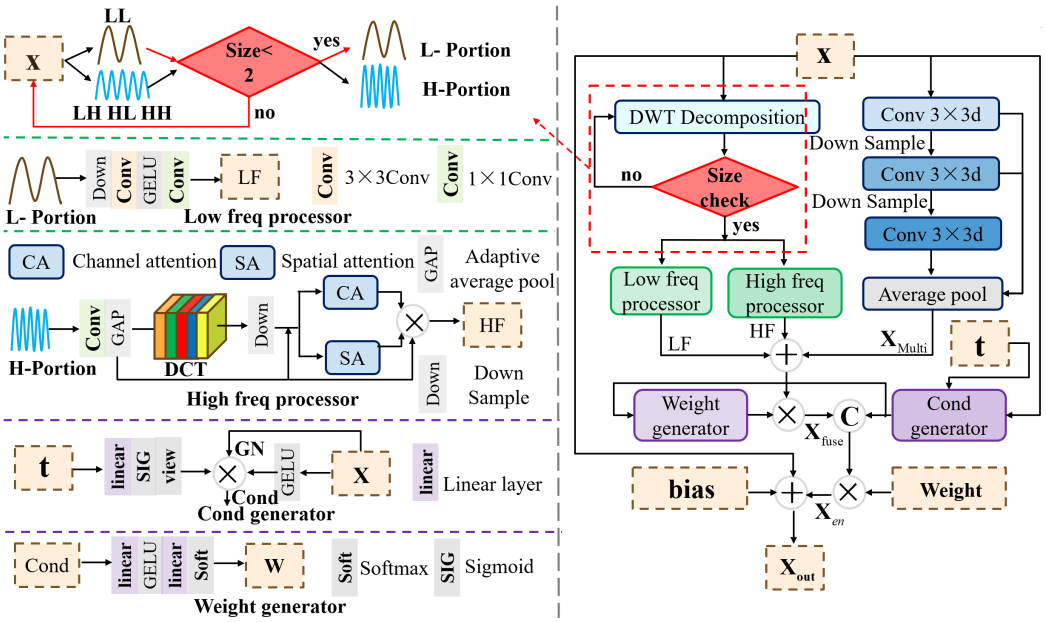


Fig. 5. Architecture diagram of MFDDFM.

accurately enhance infrared high-frequency details. Finally, a conditional generator and a weight generator are introduced to generate dynamic weights by integrating temporal information; these weights are then used for adaptive fusion of the processed low-frequency and high-frequency features. This allows the module to dynamically adjust the fusion ratio of multiscale features according to different environments (e.g., hazy weather, nighttime lighting, daytime), ultimately outputting coordinated and consistent multiscale features. Through the full-process design of decomposition, dedicated processing, and dynamic fusion, this module effectively resolves multiscale frequency representation defects and provides precise frequency-domain support for the generation of multiscale features in the conversion from visible light-to-infrared images.

The MFDDFM is embedded between each layer of the input layer of the denoising network, so as to realize the decomposition of multiscale frequencies and dynamic fusion based on time and features.

As shown in Fig. 5, first, the input feature X is decomposed into a low-frequency component (LL) and high-frequency components (HH, HL, and LH) via DWT. A judgment is made based on the size of the low-frequency component LL to determine whether to perform an iterative calculation. If required, DWT transformation is continued on the low-frequency component LL, while the high-frequency components HH, HL, and LH are stored in the high-frequency component set. This process is repeated cyclically until the size requirement is met, thereby achieving effective decomposition of frequencies at all scales of the denoising network. The mathematical expression of DWT is shown in the, (27), at the bottom of the page.

Among them, i and j are the horizontal and vertical coordinates of x.

The output low-frequency components and high-frequency components are processed by the low-frequency processor and high-frequency processor, respectively, to achieve frequency-domain enhancement of low frequencies and high frequencies. Particularly for high-frequency components with abundant feature information, we adopt grouped convolution based on DCT to extract frequency information from the features of each channel, realizing channel-independent frequency-domain analysis and enhancing important frequency components. Subsequently, the extracted features are added to the original high-frequency components with weights to achieve the fusion of space and frequency. The combined features are input into the channel attention mechanism and spatial attention mechanism, respectively, to extract temporal channel weights and spatial weights, which suppresses irrelevant frequency bands, avoid the loss of high-frequency details, and enhance the representation of high-frequency features. The mathematical expression for low-frequency component

$$\begin{cases} \mathrm{LL} = \dfrac{\mathrm{x}(2\mathrm{i},2\mathrm{j}) + \mathrm{x}(2\mathrm{i},2\mathrm{j}+1) + \mathrm{x}(2\mathrm{i}+1,2\mathrm{j}) + \mathrm{x}(2\mathrm{i}+1,2\mathrm{j}+1)}{4} \\ \mathrm{LH} = \dfrac{\mathrm{x}(2\mathrm{i},2\mathrm{j}) - \mathrm{x}(2\mathrm{i},2\mathrm{j}+1) + \mathrm{x}(2\mathrm{i}+1,2\mathrm{j}) - \mathrm{x}(2\mathrm{i}+1,2\mathrm{j}+1)}{2} \\ \mathrm{HL} = \dfrac{\mathrm{x}(2\mathrm{i},2\mathrm{j}) + \mathrm{x}(2\mathrm{i},2\mathrm{j}+1) - \mathrm{x}(2\mathrm{i}+1,2\mathrm{j}) - \mathrm{x}(2\mathrm{i}+1,2\mathrm{j}+1)}{2} \\ \mathrm{HH} = \mathrm{x}(2\mathrm{i},2\mathrm{j}) - \mathrm{x}(2\mathrm{i},2\mathrm{j}+1) - \mathrm{x}(2\mathrm{i}+1,2\mathrm{j}) + \mathrm{x}(2\mathrm{i}+1,2\mathrm{j}+1) \end{cases} \tag{27}$$

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processing is shown in the following equation:

$$\mathrm{LF} = Conv_{1\times1}\left(GELU\left(Conv_{3\times3}\left(Downsample\left(X_{L-portion}\right)\right)\right)\right). \tag{28}$$

The mathematical expressions for high-frequency components are shown in the following equations:

$$\begin{cases} Conv_{DCT}(u,v) = \sum_{k=1}^{8}\sum_{l=1}^{8}\cos\left(\pi\left(xx+1\right)k/16\right) \\ \cos\left(\pi\left(2yy+1\right)l/16\right) \\ \mathrm{X_{HF}} = GAP\left(\left(Conv_{1\times1}\left(\mathrm{X_{H\text{-}portion}}\right)\right)\right) \\ \mathrm{X_{union}} = \eta * Conv_{DCT}\left(\mathrm{X_{HF}}\right) + \mathrm{X_{HF}} \end{cases} \tag{29}$$

$$\begin{cases} w_s = \sigma\left(Conv_{7\times7}\left(\mathrm{X_{union}}\right)\right) \\ w_c = \sigma\left(Conv_{1\times1}\left(GELU\left(Conv_{1\times1}\left(GAP\left(\mathrm{X_{union}}\right)\right)\right)\right)\right) \\ \mathrm{HF} = Downsample\left(Conv_{1\times1}\left(\mathrm{X_{H\text{-}portion}}\right) * w_c * w_s\right). \end{cases} \tag{30}$$

Among them, (29) demonstrates the initial mathematical design of DCT grouped convolution and the generation process of the combined feature $\mathrm{X_{union}}$. The DCT grouped convolution is designed into 64 groups according to the input channel count of 64, ensuring that each channel can undergo independent frequency-domain analysis. Meanwhile, a judgment option is added outside the equation, which allows adjustments to the DCT grouped convolution based on changes in channels. The input $\mathrm{X_{H\text{-}portion}}$ undergoes 3×3 convolution and *GAP* (adaptive average pooling) operations to extract the high-frequency feature $\mathrm{X_{HF}}$. $\mathrm{X_{HF}}$ is then input into the DCT grouped convolution, and added to the high-frequency feature $\mathrm{X_{HF}}$. The weight of the frequency-domain contribution is controlled by the trainable parameter η, ultimately outputting the combined feature $\mathrm{X_{union}}$.

Equation (30) illustrates the process of the spatial-channel attention mechanism processing the combined feature $\mathrm{X_{union}}$ and performing adaptive weighting on the input $\mathrm{X_{H\text{-}portion}}$. For spatial attention, we adopt a large-scale 7×7 convolution to capture broader spatial relationships, and generate spatial attention weights w_s by combining with the sigmoid function. In channel attention, we use a combination of global average pooling, two 1×1 convolutions, the *GELU* function, and the sigmoid function to generate channel weights w_c. Channel-spatial collaborative enhancement is achieved through the multiplication of these two types of weights with $\mathrm{X_{H\text{-}portion}}$ after dimension transformation. Finally, the final feature HF is output after downsampling.

On the other hand, to obtain feature maps at all scales, we perform two rounds of downsampling on the input feature X and use 3×3 convolution to extract spatial-domain feature information at various scales. After channel concatenation, the multiscale features undergo average pooling and output X_{Multi}.

To achieve dynamic fusion of time-based multiscale feature information, we design a conditional generator that performs mapping based on the temporal vector and input feature S to generate conditional gating weights. Its mathematical expression is shown in the following equation:

$$\mathrm{cond} = view\left(\sigma\left(linear(\mathrm{t})\right)\right) \cdot GELU(\mathrm{X}) \cdot GN(\mathrm{X}). \tag{31}$$

We map the input temporal vector using a linear layer, activate it with the sigmoid function, and reshape it into the shape (B, C, 1, 1) via *view* function—laying the foundation for dimension matching with the input feature X. For the input feature X, we first perform activation via *GELU* function and adaptive group normalization via *GN*, respectively, then multiply it with the temporal vector gating weights. Finally, we achieve the generation of conditional gating weights through the integrated approach of normalization, gating, and activation.

On the basis of the conditional generator, to achieve effective fusion of conditional gating weights and multiscale features, we design a weight generator. It utilizes a combination of linear layers, *GELU* function, and *Softmax* function to realize the nonlinear mapping of conditional gating weights to the dimension of multiscale features. Its mathematical expression is shown in the following equation:

$$\mathrm{W} = Soft\max\left(linear\left(GELU\left(linear\left(\mathrm{cond}\right)\right)\right)\right). \tag{32}$$

The output conditional gating weights are multiplied by the multiscale features, outputting the feature $\mathrm{X_{fuse}}$ and realizing the dynamic fusion of time- and feature-based multiscale information.

To solve the problem of feature solidification in residual networks and achieve adaptive modulation of residuals, we introduce learnable weights and bias terms into the residual connection network. The learnable weights are used to control the intensity of feature enhancement, while the learnable biases regulate feature offset, thereby realizing dynamic adjustment and accurate matching of features. Its mathematical expression is shown in the following equation:

$$\mathrm{X_{out}} = \mathrm{X} + \mathrm{X_{en}} \cdot \mathrm{Weight} + \mathrm{bias}. \tag{33}$$

D. Multiscale Enhancement Loss Function

In the task of generating infrared images from aerial visible light, traditional loss functions mostly focus on a single scale or semantic level. They fail to adapt to the multiscale feature requirements of large-scale building structures, medium-scale vehicle thermal distributions, and small-scale hot spot details—leading to severe loss of multiscale information during gradient propagation and significant deviations in the structural integrity and thermal feature authenticity of generated images. To fully adapt the network's gradient loss to the multiscale characteristics of aerial images, and form a forward–backward multiscale information closed loop with the previously proposed multiscale network module and noise scheduling optimization (SOBOME), we ensure multiscale consistency throughout the entire link from diffusion process and network feature extraction to loss constraint. Accordingly, we propose a multiscale enhancement loss function. This loss function takes the multiscale loss of high, medium, and low-frequency information combined with LPPIS perceptual loss as its core. Targeting the frequency band characteristics of infrared images, the multiscale loss designs dedicated

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loss terms for high, medium, and low-frequency features, respectively, precisely constraining the generation accuracy of features at each scale to ensure that large-scale structures are not distorted, medium-scale thermal distributions are not confused, and small-scale details are not lost. On the semantic level, the LPPIS perceptual loss perceives the authenticity of infrared thermal distributions based on deep learning models, making up for the deficiency of traditional losses in semantic consistency. By weighted fusion of multiband loss and perceptual loss, gradients can simultaneously enhance pixel-level multiscale matching and semantic-level multiscale authenticity during backpropagation. Ultimately, this achieves closed-loop optimization of multiscale features from generation to constraint, enabling the model to not only ensure the pixel accuracy of structure and details but also meet the perceptual authenticity of infrared physics and semantics in infrared image generation under aerial multiscale scenarios. The mathematical expression of the loss function is shown in the following equation:

$$\mathrm{L} = c_{\varepsilon_t}\left\| m_t^{GETP}(y - x_0) + \sqrt{\zeta_t}\varepsilon - \varepsilon_\theta(x_t, t)\right\| + \lambda\mathcal{L}_{\mathrm{MLOG}}. \quad (34)$$

Among them, m_t^{GETP} is the diffusion coefficient of this study at time-step t, ζ_t is the variance of this study at t, $\mathrm{L}_{\mathrm{MLOG}}$ is a functional auxiliary term, and λ is a hyperparameter.

Equation (35) is the mathematical expression of the auxiliary term

$$\mathrm{L}_{\mathrm{MLOG}} = \lambda_1 X_{FH} + \lambda_2 X_{MH} + \lambda_3 X_{LH} + \lambda_4 X_{LPIPS}. \quad (35)$$

In the equation, λ_1, λ_2, λ_3, and λ_4 are hyperparameters. X_{FH}, X_{MH}, and X_{LH} are feature differences of high, medium, and low frequencies at three scales between the real infrared image and the generated infrared image, extracted by Gaussian factors. X_{LPIPS} is the LPPIS perceptual loss between the real infrared image and the generated infrared image at the original scale, which improves the perceptual effect of image generation. The three feature differences X_{FH}, X_{MH}, and X_{LH} are combinations of feature differences at the original scale, 0.75 scale, and 0.5 scale, realizing gradient constraint on the multiscale features of the generated image.

IV. Results

A. Experimental Settings

1) Model and Hyperparameters: Our framework consists of two parts: one is a pretrained VQGAN serving as the encoder and decoder for the latent space, and the other is a diffusion bridge model. During the actual training phase, we first perform basic preprocessing on images: normalize pixel values to [0, 1] to eliminate brightness differences, and standardize image sizes to match the model input requirements. Subsequently, the images are input into the Latent Space via the Encoder in the pretrained VQGAN, which compresses high-dimensional pixel data ($256 \times 256 \times 3$) into low-dimensional latent vectors (embedding dimension 3). The 8192-codebook not only reduces computational complexity but also avoids detail loss in traditional downsampling, providing efficient and complete basic features for subsequent feature extraction. In the training phase, the number of time steps for the diffusion bridge is set to 1000, while it is set to 200 during inference. For fairness, all methods in the comparative experiments are trained for 100 epochs, using vGPU-32 GB and 24vCPU AMD EPYC 7T83 64-Core Processor as training hardware. Meanwhile, to ensure the validity of the evaluation, each experiment of different methods in the comparative experiments is repeated three times, and the comparison results are presented as the mean plus standard deviation of the three repetitions.

2) Evaluation: To evaluate the quality and distribution consistency of the generated images, we use peak SNR (PSNR) and structural similarity (SSIM) to measure the similarity between the generated images and the original images in terms of pixels and structure. The higher the value, the better the similarity. Image similarity is evaluated based on a deep learning model using learning-perceived image block similarity (LPIPS), and the lower the value, the more similar the perception. The distance between the distribution of the generated image and the distribution of the real image is evaluated by the Frechet inception distance (FID). The lower the value, the better the generation consistency.

3) Dataset and Baseline: To demonstrate the ability to convert visible light images to infrared images from an aerial perspective, this article uses the publicly available dataset DroneVehicle as the dataset for this article. The infrared images in the DroneVehicle dataset [40] were collected from the FLIR Vue Pro R thermal imaging sensor carried by the DJI Mavic 2 Enterprise advanced drone. Its infrared band is 7.5–13.5 μm, which can reflect the thermal radiation characteristics of the object. It is suitable for all-weather monitoring tasks. The wavelength range of visible light (RGB) images is 400–700 nm, mainly capturing the reflection characteristics of objects to visible light. There are 17 990 pairs of paired images in the training set, 4498 pairs in the validation set, and 4498 pairs in the test set. To verify the effectiveness of our method at various scales, we selected 500 images each from a single-target scene, a dense-target scene, and a multiscale mixed-target scene in the original test set to evaluate the effect of the method. The comparison methods include EGG-M [11], EGG-U [11], BBDM-VQ4 [21], BBDM-VQ8 [21], PIX2PIX [9], CycleGAN [10], ACLGAN [29], DRIT [30], DRAVIT [12], MUNIT [28], DSMAP [32], UNSB [22], and StegoGAN [13].

B. Ablation Experiment

It can be seen from the comparison of ablation experiments in Table I that the reconstruction of the diffusion process differential equation improves the diffusion process's ability to generate multiscale infrared images, achieving initial model optimization. The introduction of the gated polynomial enhancement term in drift and the thermodynamic factor in the diffusion term—through optimizing time-step scheduling and dynamic variance adjustment—results in an increase of 0.13 in PSNR, 1.06 in SSIM, and a decrease of 1.26 in FID. Both SSIM and generation distribution are improved. However, since the SOBOME diffusion process optimizes scale generation from a mathematical perspective without enhancing network feature extraction or perceptual constraints,

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TABLE I
COMPARISON OF ABLATION EXPERIMENTS

SOBOME	MENBOTM	MFDDFM	MELF	PSNR	SSIM	LPIPS	FID
				14.83	48.22	31.65	49.48
√				14.96	49.28	31.61	48.22
√	√			14.98	49.67	31.93	44.71
√	√	√		15.03	49.71	31.52	44.13
√	√	√	√	15.06	49.78	30.71	43.88

LPIPS only decreases by 0.04, and semantic feature perception remains limited.

After adding MENBOTM, FID drops significantly by 4.49, indicating that dynamically modulating infrared multiscale information through time effectively enhances the authenticity of infrared image generation and achieves substantial optimization of the consistency of generated image distribution. Meanwhile, SSIM, PSNR, and LPIPS all improve to varying degrees, demonstrating that the combination of temporal modulation and multiscale branches further optimizes the structural integrity of images.

The introduction of MFDDFM increases PSNR from 14.98 to 15.03 and SSIM from 49.67 to 49.71, showing that the module's frequency-domain decomposition and dynamic fusion effectively repair high-frequency details and further improve image fineness and structural matching. At the same time, LPIPS decreases to 31.52 and FID to 44.13; however, LPIPS rises to 32.62. The perceptual deviation caused by multiscale feature redundancy is further resolved, and the offset of local scale feature distribution is reduced.

The integration of MELF forms bidirectional optimization with SOBOME's diffusion optimization, MENBOTM's feature enhancement, and MFDDFM's frequency-domain processing. The multiscale loss refines the generation error of features at each scale, leading to slight improvements in PSNR and SSIM. The LPIPS perception effectively enhances the infrared perceptual authenticity of generated images, with LPIPS decreasing by 0.81 and FID by 0.25. Closed-loop constraints make the distribution of multiscale features more consistent with the forward process of real infrared images.

Through the progressive synergy of diffusion optimization → feature enhancement → frequency-domain repair → loss constraint, the above four innovative submethods continuously increase PSNR and SSIM while decreasing LPIPS and FID, gradually solving the multiscale generation problem in converting aerial visible light-to-infrared images.

C. Ablation Experiment

Table II focuses on the multiscale visible-to-infrared image generation task in aerial scenarios. We conduct a horizontal comparison of the core performance indicators of 14 mainstream methods, covering GAN series and diffusion model series, quantitatively verifying the effectiveness of our method in this task.

TABLE II
IN THE COMPREHENSIVE SCENARIOS, THE COMPARISON OF EXPERIMENTAL INDICATOR RESULTS WITH MAINSTREAM METHODS IS PRESENTED BELOW. BOLD VALUES INDICATE THE BEST RESULTS, WHILE RED VALUES INDICATE THE SECOND-BEST. DATA ARE EXPRESSED AS THE MEAN ± STANDARD DEVIATION OF THREE EXPERIMENTAL REPETITIONS

Method	Reference	PSNR	SSIM	LPIPS	FID
PIX2PIX	CVPR2017	9.4± 0.35	3.6± 0.71	69.3± 2.15	168.35 ±4.56
CycleGAN	ICCV2017	12.94± 0.25	27.36 ±0.64	34.90± 1.17	129.35 ±3.54
DRIT	ECCV2018	10.75± 0.29	5.85± 0.65	37.43± 1.05	50.69± 1.57
MUNIT	ECCV2018	12.70± 0.23	31.45 ±0.60	34.35± 1.10	51.91± 1.62
ACLGAN	ECCV2020	8.43± 0.32	14.68 ±0.78	69.22± 2.30	168.58 ±4.60
DSMAP	ECCV2020	12.39± 0.26	30.45 ±0.63	39.78± 1.28	70.66± 2.45
EGGAN-M	ICRA2023	14.36± 0.18	48.86 ±0.52	43.80± 1.25	84.55± 2.82
BBDM-VQ4	CVPR2023	14.83± 0.15	48.22 ±0.48	31.65± 0.93	49.48± 1.65
BBDM-VQ8	CVPR2023	13.79± 0.22	35.66 ±0.65	41.03± 1.43	107.32 ±3.27
EGGAN-U	ICRA2023	10.85± 0.30	8.72± 0.85	42.20± 1.56	91.68± 3.13
UNSB	ICLR2024	13.54± 0.24	25.88 ±0.62	37.06± 1.23	65.75± 2.23
StegoGAN	CVPR2024	12.33± 0.28	22.91 ±0.78	41.52± 1.34	119.84 ±3.85
DRAVIT	TGRS2024	12.99± 0.20	25.13 ±0.55	38.22± 1.15	49.47± 1.55
Ours	**Ours**	**15.06± 0.12**	**49.78 ±0.45**	**30.71± 0.88**	**43.88± 1.35**

First, in terms of the two pixel-level indicators—PSNR (measuring pixel similarity between generated and real infrared images) and SSIM (evaluating structural consistency)—our method achieves the highest values. By leveraging SOBOME, MELF, and MFDDFM, we enhance the nonlinear mapping of features at different scales and avoid interference between high and low-frequency features. This results in a PSNR of 15.06 ± 0.12, an increase of 1.55% compared to the second-best method BDM-VQ4, enabling precise multiscale pixel-level control. Meanwhile, the SSIM reaches 49.78 ± 0.45, a 1.88% improvement over the second-best method EGGAN-M. This effectively addresses the mode collapse of EGGAN-M and its insufficient adaptation to multiscales in aerial perspectives, achieving stable multiscale structural consistency.

In terms of LPIPS (assessing the semantic authenticity of generated images) and FID (measuring the rationality of generation distribution), our method exhibits significant advantages. Our method achieves the lowest LPIPS value of 30.71 ± 0.88, a significant decrease of 9.4% compared to the second-best BBDM-VQ4. By enabling the model to perceive the authenticity of multiscale semantic levels, we avoid visible light texture residues and perceptual errors in thermal feature distortion, making the generated images more consistent with human cognitive logic of infrared images. For the FID indicator, our method achieves the lowest value of 43.88 ± 1.35. Through the synergy of the four innovative methods, the multiscale

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TABLE III

IN SINGLE-TARGET SCENARIOS, THE COMPARISON OF EXPERIMENTAL INDICATOR RESULTS WITH MAINSTREAM METHODS IS PRESENTED BELOW. BOLD VALUES INDICATE THE BEST RESULTS, WHILE RED VALUES INDICATE THE SECOND-BEST

Method	Reference	PSNR	SSIM	LPIPS	FID
DRIT	ECCV2018	11.28	8.13	35.51	53.50
EGGAN-M	ICRA2023	13.87	42.01	44.73	116.90
BBDM-VQ4	CVPR2023	16.54	57.83	27.64	59.96
UNSB	ICLR2024	13.66	26.62	26.84	82.37
DRAVIT	TGRS2024	13.72	23.49	37.05	60.97
Ours	**Ours**	**16.78**	**59.48**	**26.59**	**55.29**

TABLE IV

COMPARISON OF EXPERIMENTAL INDEX RESULTS WITH MAINSTREAM METHODS IN DENSE-TARGET SCENARIOS SHOWS THAT THE BOLD ONES ARE THE BEST, WHILE THE RED ONES ARE SUBOPTIMAL

Method	Reference	PSNR	SSIM	LPIPS	FID
DRIT	ECCV2018	10.60	4.14	36.81	60.95
EGGAN-M	ICRA2023	13.06	28.56	45.46	134.37
BBDM-VQ4	CVPR2023	15.06	49.67	29.22	42.66
UNSB	ICLR2024	13.21	18.55	37.42	103.99
DRAVIT	TGRS2024	12.96	23.25	38.48	178.70
Ours	**Ours**	**15.27**	**51.32**	**28.12**	**34.67**

feature distribution of generated images is more aligned with that of the real infrared image dataset, significantly reducing feature offset at all scales and achieving reliable generation distribution consistency.

In summary, the indicator comparison in Table II clearly shows that our proposed multiscale enhancement-based method, through end-to-end optimization of multiscale issues in the aerial visible-to-infrared image generation task, achieves optimal matching of pixel-level indicators and addresses the key pain points of semantic perception and generation consistency.

Single-target scenarios are typically characterized by a single type of core target and multiscale details, featuring high homogeneity and high precision requirements for details. They require generation models to deepen the detailed expression of core targets, and can be used to test the generalization ability of generation models in complex scenarios and verify the effectiveness of multiscale enhancement methods. Table III presents a comparison of indicator data between our method and typical methods in single-target scenarios. It can be seen that our method leads comprehensively in the four core indicators (PSNR, SSIM, LPIPS, and FID), with particularly significant improvements of 3.8% and 7.8% in LPIPS and FID, respectively.

Meanwhile, there are certain differences in the comprehensive data between Tables III and II, attributed to the more concentrated feature distribution and higher homogeneity in single-target scenarios. Therefore, our method achieves significant improvements in three indicators (PSNR, SSIM, and LPIPS) compared to the comprehensive scenario, with the SSIM indicator increasing by nearly ten points. However, the model needs to achieve a fine distinction in a high-similarity feature space. The thermal distribution differences between different vehicle models impose higher requirements on the precision of distribution fitting, leading to a slight increase in the FID indicator.

Fig. 6 shows qualitative comparisons of different methods in single-target scenarios. From left to right, the images correspond to low-light, daytime, night-light, daytime, and daytime conditions. In low-light/night-light images, our method optimizes the generation rhythm of multiscale features: accelerating target contour generation in the early stage and refining small-scale details later, which resolves blurred contours and missing details in methods like UNSB and DRIT. In daytime images, our method can effectively separate high-frequency and low-frequency features. It not only preserves vehicle structures, thermal distributions, and details, but also maintains the large-scale contours of roads and green belts. Particularly in the comparison between the second image and the ground truth (GT), the vehicle shape and road texture generated by our method show almost no difference from the GT. This addresses the thermal distribution distortion issue existing in mainstream methods, further verifying the accuracy of our method in generating multiscale details for single targets.

Dense-target scenarios are typically characterized by dense overlap of multiple target types and high interweaving of multiscale features, with many target categories and strong scale confusion. They require generation models to accurately distinguish multiscale features in dense distributions and efficiently fuse multiscale details, serving to test models' ability to decouple and integrate multiscale features in complex scenarios, further verify the effectiveness of multiscale enhancement methods in handling highly complex feature interaction scenarios, and complement single-target scenarios in performance—jointly highlighting the method's generalization and multiscale modeling advantages across scenarios.

Due to the high adaptability between the multiscale overlap of dense-target scenarios and the proposed multiscale enhancement mechanism, as shown in Table IV, our method achieves full-dimensional optimization in four indicators (PSNR: 15.27, SSIM: 51.32, LPIPS: 28.12, and FID: 34.67) in dense-target scenarios compared to comprehensive scenarios. It effectively meets the requirements of high multiscale feature distinguishability and fusion accuracy in such scenarios. Compared with the second-best method, BBDM-VQ4, the FID indicator of our method shows the most significant improvement (a 18.73% reduction), specifically addressing the confusion, loss, and distortion of dense multiscale features.

Multiscale mixed-target scenarios are typically featured by the coexistence of multiple heterogeneous target types and cross-category interweaving of multiscale features, with characteristics of target heterogeneity, cross-category scale differences, and complex environmental interference. They require generation models to accurately decouple multiscale features among multiple target types and efficiently fuse cross-category details in complex environments—serving to test models' ability to decouple and fuse multiscale cross-category features in ultracomplex scenarios, further

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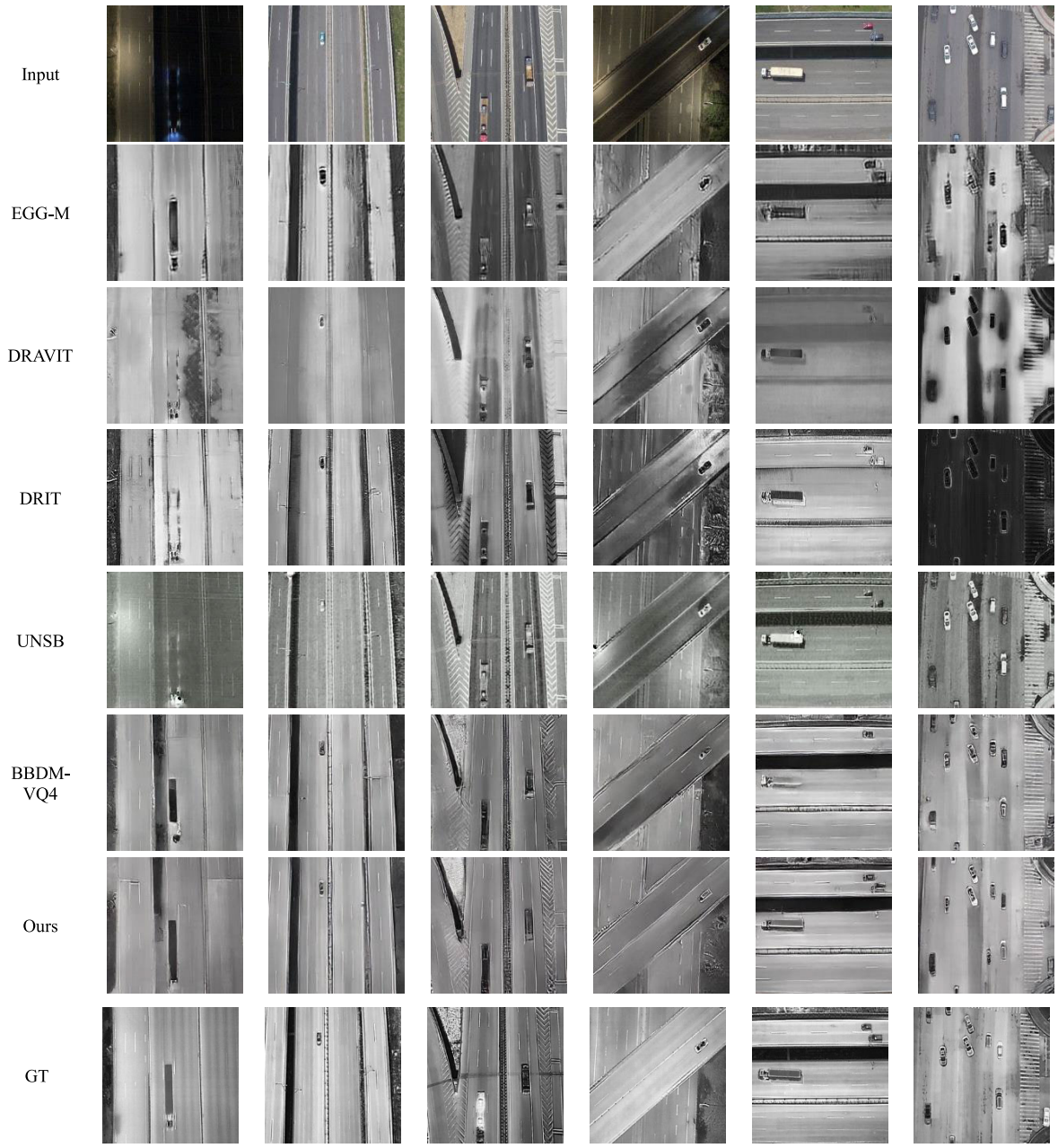


Fig. 6. Comparison of experimental qualitative results with mainstream methods in a single-target scenario.

verify the effectiveness of multiscale enhancement methods in addressing the combined challenges of multiple targets, multiscale differences, and environmental interference, and complement single-target and dense-target scenarios in performance, jointly highlighting the method's generalization and multiscale modeling advantages across scenario dimensions.

As shown in Table V, due to the superposition of three complexities (multiple targets, multiscale differences, and complex environments) in multiscale mixed-target scenarios, the proposed method's PSNR, SSIM, LPIPS, and FID indicators show a certain degree of decline compared with those in the comprehensive scenario (see Table II). Particularly, FID increases by 18.31 points, as higher confusion of multiscale features further raises the difficulty of feature extraction and generation. However, the proposed multiscale enhancement method still outperforms the second-best method: PSNR,

TABLE V

COMPARISON OF EXPERIMENTAL INDEX RESULTS WITH MAINSTREAM METHODS IN MULTISCALE MIXED-TARGET SCENARIOS SHOWS THAT THE BOLD ONES ARE THE BEST, WHILE THE RED ONES ARE SUBOPTIMAL

Method	Reference	PSNR	SSIM	LPIPS	FID
DRIT	ECCV2018	9.93	7.56	41.58	79.32
EGGAN-M	ICRA2023	12.97	46.63	44.31	111.95
BBDM-VQ4	CVPR2023	13.67	45.24	35.61	74.19
UNSB	ICLR2024	13.18	28.57	39.85	77.52
DRAVIT	TGRS2024	12.05	21.60	42.31	83.37
Ours	**Ours**	**13.75**	**47.99**	**34.88**	**62.79**

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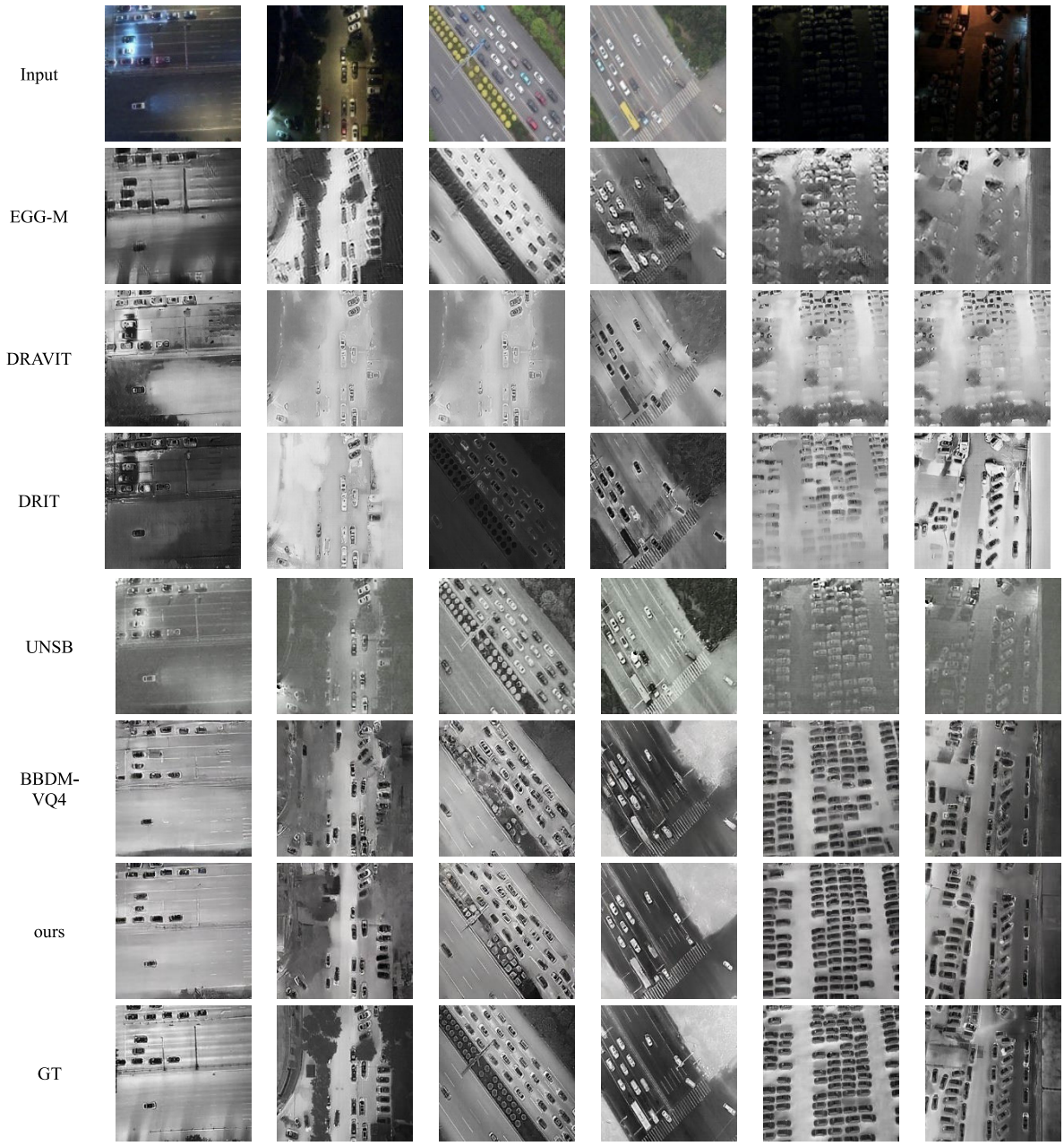


Fig. 7. Comparison of experimental qualitative results with mainstream methods in dense-target scenarios.

SSIM, LPIPS, and FID are improved by 0.585%, 6.08%, 2.05%, and 15.36%, respectively. This maintains accuracy in scenarios with pixel confusion of multiple targets, enhances structural distinguishability among targets, and makes the distribution of multiple targets more consistent with real infrared images.

Figs. 7 and 8 present qualitative comparisons of different methods in dense-target scenarios and multiscale mixed-target scenarios, respectively. For Fig. 7, from left to right, the images correspond to night-light, nightlight, daytime, daytime, low-light, and low-light conditions. For Fig. 8, from left to right, the images correspond to night-light, night-light, daytime, hazy, hazy, and hazy conditions.

In hazy scenarios, to address the overall blurriness of comparison methods, our method separates high-/low-frequency haze noise from target features—it preserves the large-scale structures of buildings and vehicles while restoring medium-scale thermal distributions and small-scale details, ensuring generated images maintain multiscale feature clarity under haze.

In night-light scenarios, targeting the false feature interference from artificial light in comparison methods, our method uses time vectors to dynamically adapt to feature generation logic under night-light, distinguishing light interference from real thermal features for more authentic thermal characteristics in generated images.

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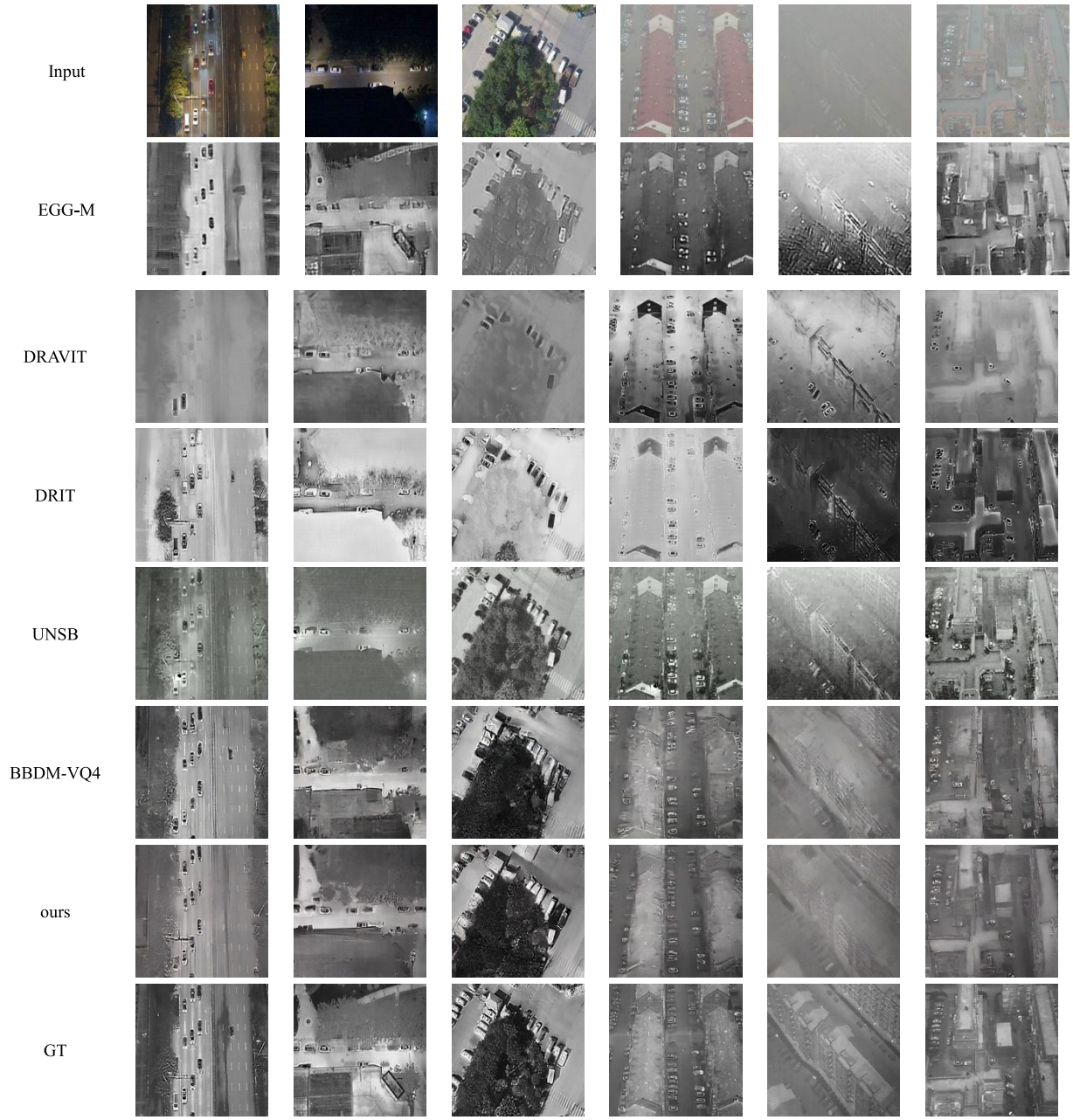


Fig. 8. Comparison of experimental qualitative results with mainstream methods in multiscale mixed-target scenarios.

In daytime scenarios, compared with methods like BBDM-VQ4 and EGG-M (which suffer from overlapping features of trees, buildings, and vehicles), our method enhances the distinction of large-scale contours among these targets and refines thermal intensity differences between trees and vehicles, making the multiscale features of generated multitype targets as consistent with GT images as possible.

V. Conclusion

This article addresses core multiscale challenges in aerial visible-to-infrared image generation; by proposing a multiscale end-to-end optimization model for complex aerial scenarios, it overcomes limitations in feature scheduling, network representation, and loss constraints, boosting scale adaptability, detail authenticity, and semantic consistency. In diffusion scheduling, a gated polynomial enhancement term and thermodynamic factor enable alignment with the "structure-first, detail-later" rule, supporting ordered multiscale infrared feature generation; the two-layer network (pretime-modulated multiscale enhancement + frequency decomposition fusion module) uses time parameters to balance large-structure integrity and small-detail authenticity, resolving high-low frequency interference and loss of high-frequency thermal details; multiscale (high-medium-low) loss combined with LPPIS perceptual loss forms a forward–backward constraint loop, avoiding information attenuation and semantic deviation. Experimental results on the DroneVehicle dataset validate its effectiveness—achieving PSNR 15.06, SSIM 49.78, LPIPS 30.71, FID 43.88, out-

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performing advanced GANs and diffusion models—and it excels in capturing small-scale hotspots, decoupling overlapping features, and distinguishing cross-category features across scenarios, demonstrating strong robustness. However, the model suffers slight performance degradation in ultracomplex scenarios, has higher computational complexity limiting real-time applications, and needs further validation for generalization across diverse datasets. Future work will focus on lightweight network design to improve inference speed, expand datasets for better generalization, and integrate environmental adaptive mechanisms to address ultracomplex scenario challenges, advancing practical aerial visible-to-infrared generation.

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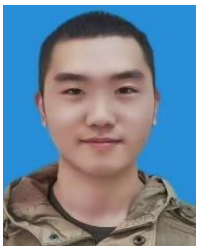
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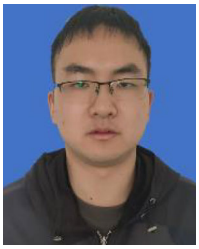
He has published several papers in IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON MULTIMEDIA, *Expert Systems with Applications* (ESWA), *Defence Technology* (DT), IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Neurocomputing*. Furthermore, he has published three monographs and eight patents have been granted. His research interests include neural networks, computer vision, image processing, and hyperspectral image clustering.

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